Problem Statement

Customer Value, Risk & Retention Analysis — Indian Retail Banking Dataset

1. Background

A retail bank maintains data on its customers, their account activity, and the branches that service them. The data is spread across three related tables: a Customers table (profile, account, credit, and risk attributes), a Branch Master table (branch, city, and bank details), and a Transactions Log (individual credit/debit transactions per customer). Leadership wants to understand where value and risk are concentrated across the customer base, which accounts require urgent attention, and which customers are disengaging — so that retention, collections, and cross-sell efforts can be targeted rather than applied uniformly.

2. Dataset Overview

Table	Description	Key Field(s)
Customers	One row per customer — demographics, account balance, credit score, loan status, engagement signals	customer_id (PK), branch_id (FK)
Branch Master	One row per branch — branch name, city, bank	branch_id (PK)
Transactions Log	One row per transaction — date, type (credit/debit), amount	transaction_id (PK), customer_id (FK)

3. Problem Statement

Given customer profile data, branch data, and transaction history, the bank needs to: (a) identify which customers represent the greatest combination of value and risk, (b) flag accounts that are dormant or approaching dormancy under RBI guidelines, (c) detect early behavioral warning signs of financial stress from transaction patterns, and (d) recommend targeted actions (retention, monitoring, or deprioritization) for each customer segment — backed by data, not assumption.

4. Key Business Questions to Answer

A. Value & Risk Segmentation

- Who are our highest-value customers, and what proportion of them are also high risk?
- How much total balance (₹) is concentrated in the High Value + High Risk segment?
- What distinguishes Low Risk customers from High Risk ones — credit score, income level, number of loans, or products held?
- Which branches or cities have the highest concentration of risky, high-value customers?

B. Engagement & Dormancy (RBI Compliance)

- What percentage of accounts qualify as Dormant under RBI's 24-month rule, and what balance do they hold?
- Which accounts are Inactive (12–24 months) and at risk of becoming dormant soon — an early-warning list?
- Is dormancy correlated with risk level or value tier? Are we losing high-value customers to inactivity?
- Which branches have the highest dormancy rates — a possible sign of local service issues?

C. Transaction Behavior & Early Risk Signals

- Which customers show negative net cash flow (spending more than they earn) — a leading indicator of stress?
- How does transaction volume/frequency differ between High Risk and Low Risk customers?
- Are there customers whose transaction activity has dropped sharply relative to their account history?
- What is the credit-to-debit ratio by customer segment, and does it predict risk level?

D. Product Holding & Cross-Sell

- Do customers with more products held show higher balances and better credit scores?
- How many customers hold only one product, and what is their combined value — the cross-sell opportunity?
- Is single-product ownership linked to higher dormancy or higher risk?

E. Branch-Level Performance

- Which branches carry the highest ₹ exposure in high-risk accounts?
- How does average credit score and balance vary by branch, city, and bank?
- Are risk and dormancy evenly spread, or concentrated in specific branches/cities?

5. What We Need to Find Out (Expected Outputs)

- A Value-Risk Matrix segmenting all customers into 6 groups (High/Low Value × Low/Medium/High Risk) with count and ₹ exposure per group.
- A prioritized action list of High Value + High Risk (and ideally also Dormant) customers, ranked by balance.
- An RBI-compliant dormancy breakdown (Active / Inactive / Dormant) with associated balance exposure.
- A set of early-warning indicators (negative cash flow, declining credit score band, approaching-dormancy window) to catch risk before it materializes.
- A branch-level risk and dormancy heatmap to identify localized problem areas.
- Clear, differentiated recommendations per segment — e.g., interest-rate relief for payment-stressed customers, relationship outreach for disengaged customers, reduced exposure for chronic high-risk customers.

6. Success Criteria

- Every customer can be placed into exactly one Value-Risk segment with a clear, documented threshold.
- Dormancy classification matches RBI's official 24-month definition, not an arbitrary cutoff.
- The highest ₹ exposure segments are clearly identified and rankable, not just customer counts.
- Recommendations are tied to root cause (credit stress, disengagement, or service issue) rather than a single blanket action.